

System Design

Case Studies

The 20% problems asked in 80% of interviews

- ◆ Master system design fundamentals
- ◆ Answer perfectly in interviews
- ◆ Covers all major system archetypes

CATEGORIES COVERED

■ Social & Media

■ Communication

■ Entertainment

■ Storage & Data

■ E-commerce

■ Developer Tools

■ Infrastructure

■ Maps & Location

■ Finance

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About This Guide

What This Is

A curated collection of 34 system design case studies covering the 20% of topics asked in 80% of senior engineering interviews. Each entry highlights core concepts, design decisions, and the trade-offs interviewers probe.

How to Use It

Work through each case study end-to-end. For every system ask: What are the functional requirements? What is the scale target? Where are the bottlenecks? What consistency vs availability trade-off applies? The linked articles provide deep dives.

Interview Framework

1. Clarify requirements & scale | 2. Define API | 3. High-level design | 4. Deep-dive bottlenecks | 5. Discuss trade-offs & failure modes. Apply this framework to all 34 systems.

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In-Depth Reference

01

Design Vending Machine

Developer Tools

[\[State Machine\]](#) [\[OOP\]](#) [\[Concurrency\]](#)

Classic OOP & state machine problem. Model states (idle, selecting, dispensing), handle coin input, item selection, and change dispensal with proper error handling.

[Read Case Study →](#)

02

Design Facebook

Social & Media

[\[News Feed\]](#) [\[Graph DB\]](#) [\[CDN\]](#)

News feed generation, social graph, post ranking algorithms, notifications, and massive-scale read-heavy architecture with multi-layer caching.

[Read Case Study →](#)

03

Design Distributed Job Scheduler

Infrastructure

[\[Cron\]](#) [\[Distributed Lock\]](#) [\[Queue\]](#)

Schedule and execute jobs reliably across distributed nodes. Handle retries, failures, at-least-once delivery, priority queues, and avoid duplicate execution.

[Read Case Study →](#)

04

Design Google Search

Infrastructure

[\[Indexing\]](#) [\[PageRank\]](#) [\[Crawler\]](#)

Web crawling, inverted index, ranking algorithms, query processing pipeline, spell correction, and serving billions of queries per day with low latency.

[Read Case Study →](#)

05

Design a Digital Wallet

Finance

[\[ACID\]](#) [\[Idempotency\]](#) [\[Ledger\]](#)

Double-entry ledger system, idempotent transactions, balance consistency, fraud detection, and regulatory compliance at scale.

[Read Case Study →](#)

06

Design Instagram

Social & Media

[\[Media Storage\]](#) [\[Feed\]](#) [\[CDN\]](#)

Photo/video upload pipeline, feed generation, follower graph, explore page ranking, and handling celebrity accounts with millions of followers.

[Read Case Study →](#)

07

Design Distributed Cache

Infrastructure

[\[LRU\]](#) [\[Consistent Hashing\]](#) [\[Replication\]](#)

Cache eviction policies, consistent hashing for distribution, replication for fault tolerance, and the hardest problem: cache invalidation.

[Read Case Study →](#)

08

Design Uber

Maps & Location

[\[Geo Index\]](#) [\[Matching\]](#) [\[Real-time\]](#)

Real-time driver location tracking, efficient geo-spatial indexing, rider-driver matching algorithm, surge pricing, and trip lifecycle management.

[Read Case Study →](#)

09

Design Distributed Key-Value Store

Storage & Data

[Raft] [Quorum] [LSM Tree]

Consensus algorithms (Raft/Paxos), quorum reads/writes, vector clocks for conflict resolution, compaction, and tunable consistency levels.

[Read Case Study →](#)

10

Design Notification Service

Communication

[Push] [Fan-out] [Templates]

Multi-channel delivery (push, SMS, email), fan-out on write vs read, user preferences, rate limiting per user, and retry with exponential backoff.

[Read Case Study →](#)

11

Design Spotify

Entertainment

[Streaming] [Recommendation] [CDN]

Audio streaming with adaptive bitrate, playlist management, recommendation engine, offline sync, and podcast vs music delivery differences.

[Read Case Study →](#)

12

Design Payment System

Finance

[2PC] [Idempotency] [PCI DSS]

Payment processing pipeline, two-phase commit, idempotency keys, reconciliation, chargebacks, and integrating with payment gateways like Stripe.

[Read Case Study →](#)

13

Design Leaderboard

Storage & Data

[Redis Sorted Set] [Ranking] [Real-time]

Real-time ranking with Redis sorted sets, score updates at scale, pagination of rankings, partitioned leaderboards, and handling ties correctly.

[Read Case Study →](#)

14

Design File Sharing (Dropbox)

Storage & Data

[Chunking] [Delta Sync] [Conflict]

File chunking and deduplication, delta sync to minimize bandwidth, conflict resolution, offline support, and sharing/permissions model.

[Read Case Study →](#)

15

Design Google Maps

Maps & Location

[Graph] [Routing] [Tiles]

Map tile rendering and caching, graph-based routing (Dijkstra/A*), real-time traffic integration, ETA calculation, and POI indexing.

[Read Case Study →](#)

16

Design Distributed Counter

Infrastructure

[CRDT] [Sharding] [Approximation]

Exact vs approximate counting, CRDT-based counters, sharded counters to avoid hot spots, and trade-offs between consistency and performance.

[Read Case Study →](#)

17

Design WhatsApp

Communication

[E2E Encryption] [WebSocket] [Presence]

End-to-end encryption, message delivery guarantees, online presence/typing indicators, group messaging fan-out, and media transfer pipeline.

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18

Design Netflix

Entertainment

[CDN] [Adaptive Bitrate] [Microservices]

Content ingestion and transcoding pipeline, adaptive bitrate streaming, CDN edge node strategy, recommendation system, and A/B testing infrastructure.

[Read Case Study →](#)

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Design TikTok

Social & Media

[Recommendation] [Video Pipeline] [Feed]

Short video processing pipeline, For You Page ML recommendation, creator monetization, live streaming, and handling viral content spikes.

[Read Case Study →](#)

20

Design Pastebin

Storage & Data

[URL Shortener] [TTL] [Storage Tiers]

Unique ID generation, custom URLs, expiry/TTL management, syntax highlighting, storage tier optimization, and abuse prevention.

[Read Case Study →](#)

21

Design Rate Limiter

Infrastructure

[Token Bucket] [Sliding Window] [Redis]

Token bucket vs leaky bucket vs sliding window algorithms, distributed rate limiting with Redis, and applying limits per user/IP/endpoint.

[Read Case Study →](#)

22

Design Amazon E-commerce

E-commerce

[Inventory] [Search] [Recommendations]

Product catalog, inventory management, search with filters/facets, cart and checkout, order management, and recommendation engine.

[Read Case Study →](#)

23

Design Google Docs

Developer Tools

[OT] [CRDT] [Collaboration]

Real-time collaborative editing using Operational Transformation or CRDTs, conflict resolution, revision history, comments, and offline sync.

[Read Case Study →](#)

24

Design Live Comments

Communication

[WebSocket] [Fan-out] [Pagination]

Real-time comment streaming via WebSockets, fan-out to thousands of viewers, pagination of historical comments, and moderation pipeline.

[Read Case Study →](#)

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Design Flight Booking System

E-commerce

[Inventory Lock] [Saga] [Search]

Seat inventory management, distributed transaction saga pattern, search with complex filters, price fluctuation, and third-party airline API integration.

[Read Case Study →](#)

26

Design CDN

Infrastructure

[Edge Cache] [Anycast] [Invalidation]

Edge PoP placement strategy, cache invalidation propagation, anycast routing, origin shield, and handling dynamic vs static content differently.

[Read Case Study →](#)

27

Design Online Code Editor

Developer Tools

[Sandbox] [WebSocket] [OT]

Secure code execution sandbox (containers/VMs), real-time collaborative editing, language support, output streaming, and preventing resource abuse.

[Read Case Study →](#)

28

Design Autocomplete

Developer Tools

[Trie] [Ranking] [Typeahead]

Trie-based prefix matching, personalized vs global suggestions, ranking by frequency/relevance, low-latency serving, and A/B testing suggestions.

[Read Case Study →](#)

29

Design DoorDash Delivery

E-commerce

[\[Geo Matching\]](#) [\[ETA\]](#) [\[Real-time\]](#)

Restaurant and dasher matching, real-time order tracking, ETA prediction with ML, zone-based batching, and surge/busy period handling.

[Read Case Study →](#)

30

Design Authentication System

Infrastructure

[\[OAuth2\]](#) [\[JWT\]](#) [\[MFA\]](#) [\[SSO\]](#)

OAuth 2.0/OIDC flows, JWT token lifecycle, refresh token rotation, MFA implementation, SSO across services, and session management at scale.

[Read Case Study →](#)

31

Design Kafka

Infrastructure

[\[Log\]](#) [\[Partitions\]](#) [\[Consumer Groups\]](#)

Append-only commit log, partition-based parallelism, consumer group offsets, replication for durability, and exactly-once semantics.

[Read Case Study →](#)

32

Design Slack

Communication

[\[WebSocket\]](#) [\[Workspace\]](#) [\[Search\]](#)

Workspace and channel model, real-time message delivery, threaded conversations, search over message history, and file sharing pipeline.

[Read Case Study →](#)

33

Design Distributed Locking

Infrastructure

[\[Fencing Token\]](#) [\[Lease\]](#) [\[Redlock\]](#)

Lease-based locks, fencing tokens to handle lock expiry edge cases, Redlock algorithm, and lock-free alternatives like CRDT for appropriate use cases.

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Design Twitter

Social & Media

[\[Fan-out\]](#) [\[Timeline\]](#) [\[Graph\]](#)

Tweet fan-out on write vs read for celebrities, home timeline generation, trending topics, search, and handling hot users with millions of followers.

[Read Case Study →](#)

Quick Reference

All 34 Systems at a Glance

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Interview Tips

1. Clarify Before Designing

Spend the first 5 minutes asking clarifying questions. How many users? Read-heavy or write-heavy? What is the SLA? Interviewers want to see structured thinking before you draw boxes.

2. Back-of-Envelope Estimation

Always estimate scale: QPS, storage, bandwidth. A system for 1M users looks very different from one for 1B. Use powers of 10: $10^3 = 1 \text{ K}$, $10^6 = 1 \text{ M}$, $10^9 = 1 \text{ B}$.

3. Start High-Level, Then Deep-Dive

Sketch the full system first — clients, load balancers, services, database, cache. The interviewer will then direct you to deep-dive the most interesting component.

4. Always Discuss Trade-offs

Never present one solution as the only option. Compare SQL vs NoSQL, push vs pull, sync vs async. Show you understand trade-offs, not just patterns.

5. Address Failure Modes

What happens when a node dies? When the DB is overloaded? When the network partitions? Discussing failure modes and recovery demonstrates senior-level thinking.

6. Mention Non-Functional Requirements

Availability (99.9% vs 99.99%), consistency (eventual vs strong), latency (p50 vs p99), and durability. These constraints drive your entire architecture.